AUTOMATIC LOCALIZATION OF HUMAN EYES IN COMPLEX BACKGROUND

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ABSTRACT

Based on geometrical facial features and image segmentation, this paper presents a novel algorithm for automatic localization of human eyes in grayscale still images with complex background. First of all, a determination criterion of eye location is established by the priori knowledge of geometrical facial features. Secondly, a range of threshold values that would separate eye blocks from others in a segmented facial image is estimated from the facial image histogram. Thirdly, with the progressive increase of the threshold by an appropriate step in that range, the size of the existing blocks in the segmented facial image will expand, some existing blocks will merge into one block, and some new blocks will emerge. Once two eye blocks appear from the segmented image, they will be detected by the determination criterion of eye location. Finally, the 2-D correlation coefficient is used as a symmetry similarity measure to check the factuality of the two detected eyes. In this way, the optimal threshold value can be automatically found based on the result of detection such that eyes can be accurately located. The experimental results demonstrate the high efficiency of the algorithm in runtime and correct localization rate.

1. INTRODUCTION

In an automatic face recognition system, the features or representation of a face are extracted automatically from an input facial image and then compared in a matching process [1,2]. Since face recognition is often based on the normalized facial images and normalizing input facial images mainly depends on eye location, the localization of human eyes in input facial images is a fundamental step in the process of face recognition [3]. A common method used for locating eyes is the deformable templates, proposed by Yuille et al [4] and improved by Xie et al. [5]. This method makes use of global information and hence improves the reliability of locating the contour of the eye. However, the template scheme is associated with problems such as slow convergence and lengthy processing time. In this paper, we present a novel approach to achieve fast, accurate eye detection that is robust to changes in illumination and backgrounds. The proposed method is based on geometrical facial features and image segmentation, and works on grayscale still images (256 gray levels).

2. A DETERMINATION CRITERION OF EYE LOCATION

Human faces have a special pattern that is usually distinct from the patterns of background objects in facial images. The grayscales of the pupil and the iris of an eye are usually lower than those of the skin near the eye and those of the white of the eye, Therefore, if we can find an appropriate threshold value to segment a facial image, the eyes can be separated from other facial features and background objects in the segmented facial image (i.e., the binary image, where, if the grayscale of a pixel is more than or equal to the threshold value, the grayscale of the pixel will be set to be 1 (white pixel); otherwise, it will be set to be 0 (black pixel) such as Fig. 1).

Fig.1 A segmented facial image with an appropriate threshold value.

The connected components (black pixels) in the segmented facial image are called a block. Morphological operations ("Majority") [6] need to be performed on the segmented facial image in order to remove some small

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blocks. All the blocks are labeled. The number and the position of each block are recorded. To locate eyes from an appropriately segmented facial image, a determination criterion of eye location needs to be established by the priori knowledge of geometrical facial features as follows:

- The distance between the geometrical centers of the two eye blocks should be within a certain range of pixel number such as from 20 pixels to 60 pixels.
- There is no other block in a certain area under each eye.
- The vertical distance difference between the geometrical centers of the two eye blocks is not more than a certain number of pixels.
- The size (the pixel number) in each eye block is limited in a certain range.
- There is no other block between the two eye blocks.
- The shape of each eye block is similar to a circular.
- Any block connected with or very close to the four edges of facial images is not an eye block.

3. ESTIMATING THE RANGE OF THE OPTIMAL THRESHOLD VALUE

The so-called optimal threshold value is the threshold value used to segment an input facial image so that the two eyes can be separated from other facial features and background objects. Suppose that the optimal threshold value is in a range from T_0 to T_{max}. Statistics show that in most facial images with complex background, T_0 can be set to be the grayscale corresponding to the first peak in the low gray area of facial image histograms, while T_{max} can usually be set to be 0.6. When pictures are taken in a fixed background and illumination condition, the range (T_0, T_{max}) can be limited in an even smaller area. Before T_0 is estimated, the facial image histogram needs to be smoothed.

3.1 Smoothing the facial image histogram

The grayscale histogram of an image, generally, is not smooth and that can result in a mistake to estimate the positions of peaks in the histogram. To solve the problem, an algorithm is developed as follows:

Step 1: Select the following Gaussian function,

$$g(x) = \frac{1}{\sqrt{2\pi}\,\sigma} \exp\left(-\frac{(x-x_0)^2}{2\sigma^2}\right) \qquad (1)$$

where $x = 1, 2, \cdots, L_g$, $x_0 = L_g/2$, L_g is the Length of the discrete Gaussian function.

Step 2: Take a convolution of $g(x)$ with a histogram $h(x)$ (its length is 256),

$$h_0(x) = h(x) \otimes g(x) \qquad (2)$$

where the length of $h_0(x)$ is $L_g + 255$.

Step 3: Let $h_s(x)$ be half of the sum of the two vectors that consist of the first 256 elements and the last 256 elements of $h_0(x)$, respectively, i.e.,

$$h_s(1:256) = \frac{1}{2}[h_0(1:256) + h_0(L_g : L_g + 255)] \qquad (3)$$

Step 4: Let $h(x) \Leftarrow h_s(x)$, and repeat step 2 and step 3 one more time.

Fig.2 displays an image histogram and its smoothed histogram by using the above algorithm.

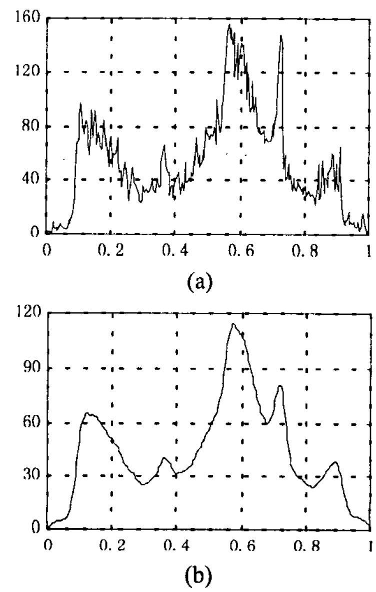

Fig.2 (a) A facial image histogram $h(x)$. (b) Its smoothed histogram $h_s(x)$.

3.2 Estimating T_0

Generally, the positions of peaks in smoothed histograms are estimated by solving the extreme values of a function. The method is usually time-consuming. Instead, here we introduce a simple way to find the position (T_0) of the first peak in the low gray area of a smoothed facial image histogram $h_s(x)$, which is described by the code of Matlab 5.3 as follows:

```
For x=0:24
hmax(x+1)=max(hs(x*10+1:(x+1)*10));
hmin(x+1)=min(hs(x*10+1:(x+1)*10));
end
x=1;
```

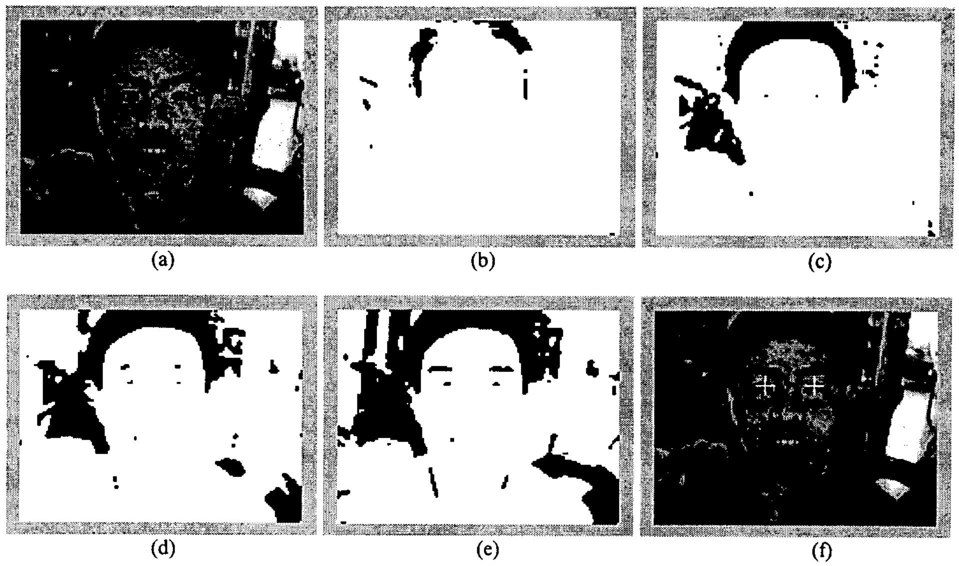

Fig. 3 Detecting and locating human eyes in segmented facial images based on automatic search for the optimal threshold value. (a) Original facial image. (b)~(e) Segmented facial images with threshold values from 0.12 to 0.3, and the threshold step is 0.06. When the threshold value became 0.3 in (e), eye blocks were detected. (f) The result of locating eyes (displayed by the while crosses)

```
while hmax(x+1)-hmax(x)>=0
x=x+1;
end
T01=find(hs==hmax(x));T0=T01(1);
```

4. DETECTING AND LOCATING HUMAN EYES

In a segmented facial image, with the progressive increase of the threshold value by a small step in the range from T_0 to $T_{\max}$, the size of the existing blocks will expand, some existing blocks will merge into one block, and some new blocks will emerge in the segmented facial image (see Fig. 3). Once two eye blocks appear from the segmented image and spread to a certain size, they will be detected by the determination criterion of eye location. However, sometimes, we found that in the detection process the two detected eyes might not be real eyes because of the interference of complex image background. Therefore, the 2-D correlation coefficient is used as a similarity measure to check the factuality of the two detected eyes.

First of all, taking out a small sub-image with the size of $M \times N$ (such as 21×21), denoted with A_L and A_R respectively for the left eye and the right eye, from the original facial grayscale image at the geometrical center of each detected eye block. Secondly, flipping the matrix A_R in left direction to get a new matrix B_R. Finally, computing the 2-D correlation coefficient r as follows:

$$r = \frac{\sum_{m=0}^{M-1}\sum_{n=0}^{N-1}[A_L(m,n)-\overline{A}_L][B_R(m,n)-\overline{B}_R]}{\sqrt{\sum_{m=0}^{M-1}\sum_{n=0}^{N-1}[A_L(m,n)-\overline{A}_L]^2 \sum_{m=0}^{M-1}\sum_{n=0}^{N-1}[B_R(m,n)-\overline{B}_R]^2}} \quad (4)$$

where $\overline{A}_L$ and $\overline{B}_R$ are respectively the mean grayscale values of A_R and B_R.

Before the process of the increase of threshold value for searching for the optimal threshold, r is set to be zero. During the process, if finding $r \geq 0.5$, the detected two eyes are usually true. If finding $r < 0.5$, the process will continue until finding $r \geq 0.5$. If finding $0.5 > r > 0$ always in the whole process, the maximum of r, denoted with $r_{\max}$, will be selected and the two detected eyes corresponding to $r_{\max}$ will be taken as true eyes. If finding $r \leq 0$ always in the whole searching process, no true eyes are detected. In this way, the optimal threshold can be automatically found such that eyes can be accurately located.

5. EXPERIMENTAL RESULTS

We used a face database (60 people, 240 (frontal) images) to evaluate the performance of the proposed

Fig. 4 Several examples for locating eyes (Eye positions are denoted with white crosses)

algorithm. The algorithm was programmed by the code of Matlab 5.3 and worked on the Pentium III/450M personal computer. The average runtime is about 2 seconds per image and the correct localization rate is close to 99%. The main reason for detection failure is that there are some patterns in facial image backgrounds that are similar to human face patterns. We also tested 28 nonfacial images. The proposed algorithm gave only two false alarms also because there is a false facial pattern in each of the two nonfacial image backgrounds. The experimental results tell us that taking facial images should, as far as possible, avoid those backgrounds containing false facial patterns. In this way, the proposed algorithm can reach high efficiency in both correct localization rate and runtime. Fig. 4 displays some examples of locating eyes using the proposed algorithm.

6. CONCLUSSIONS

By automatic search for the optimal threshold value, a simple and novel algorithm is developed in this paper for locating human eyes based on the determination criterion of eye location and the similarity measure of two eyes. The experimental results showed the proposed algorithm is robust to changes in illumination and backgrounds and demonstrated the high efficiency in runtime and correct localization rate. In our future research, we shall use the facial skin color information to reduce the affection of false facial patterns in backgrounds.

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